

TMA4315 Generalized linear models H2018

Module 9: SUMMING UP

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(Latest changes: 22.11 second version).

Overview

Topics

- ▶ course content and learning outcome
- ▶ reading list
- ▶ course topic and modules
 - ▶ core concepts: exponential family, models: LM/GLM/mvGLM/LMM/GLMM, likelihood, maximum likelihood, score vector, Fisher information, Fisher scoring, Wald/LRT(/score) tests, deviance, AIC
 - ▶ incoming questions: overview of models (including categorical regression), exponential family and canonical link (why?), likelihood-score-Fisher information, which tests for what, model assessment (deviance and residuals)
- ▶ exam and exam preparation
- ▶ suggestions for statistics-related courses in year 4 and 5
- ▶ questionnaire

About the course

Content

- ▶ Univariate exponential family.
- ▶ Multiple linear regression.
- ▶ Logistic regression.
- ▶ Poisson regression.
- ▶ models for multinomial data,
- ▶ General formulation for generalised linear models with canonical link.
- ▶ Likelihood-based inference with score function and expected Fisher information.
- ▶ Deviance. AIC. Wald and likelihood-ratio test.
- ▶ Linear mixed effects models with random components of general structure.
- ▶ Random intercept and random slope.
- ▶ Generalised linear mixed effects models.

More Content

- ▶ Strong emphasis on programming in R.
 - ▶ you will not be asked to code in the exam, but should understand the code, and the output

Possible extensions:

- ▶ quasi-likelihood,
- ▶ over-dispersion (covered a bit),
- ▶ quantile regression.

Learning outcome

Knowledge

The student can assess whether a generalised linear model can be used in a given situation and can further carry out and evaluate such a statistical analysis. The student has substantial theoretical knowledge of generalised linear models and associated inference and evaluation methods. This includes regression models for normal data, logistic regression for binary data and Poisson regression.

The student has theoretical knowledge about linear mixed models and generalised linear mixed effects models, both concerning model assumptions, inference and evaluation of the models. Main emphasis is on normal, binomial and Poisson models with random intercept and random slope.

Skills

The student can assess whether a generalised linear model or a generalised linear mixed model can be used in a given situation, and can further carry out and evaluate such a statistical analysis.

Final reading list

Fahrmeir, Kneib, Lang and Marx (2013): Regression, Springer: eBook (free for NTNU students).

<https://link.springer.com/book/10.1007%2F978-3-642-34333-9>

- ▶ Chapter 2: 2.1, 2.2, 2.3, 2.4, 2.10
- ▶ Chapter 3 (also on reading list for TMA4267)
- ▶ Chapter 5: 5.1, 5.2, 5.3, 5.4, 5.8.2
- ▶ Chapter 6: but not p 344-345 nominal models and latent utility models, not 6.3.2 Sequential model, and not category specific variables on page 344-345.
- ▶ Chapter 7: 7.1, 7.2, 7.3, 7.5, 7.7, 7.8.2. [In greater detail: pages 349-354 (not "Alternative view on the random intercept model"), 356-365 (not 7.1.5 "Stochastic Covariates""), 368-377 (not "Bayesian Covariance Matrix""), 379-380 (not "Testing Random Effects or Variance Parameters" "", only last part on page 383), 383 (middle), 389-394, 401-409. Note: Bayesian solutions not on the reading list.]
- ▶ Appendix B.1, B.2, B.3 (not B.3.4 and B.3.5), B.4

In addition to the Fahrmeir et al book, on the reading list is also:

- ▶ All the 9 module pages (but module 1 and 9 does not have theory that is not in 2-8).
- ▶ The three compulsory exercises.

The modules

1. Introduction (exponential family, Rstudio, ggplot and R Markdown)
2. Multiple linear regression (emphasis on likelihood)
3. Binary regression (independent responses, binary individual and grouped response)
4. Count and continuous positive response data (independent responses, Poisson- and gamma regression)
5. Generalized linear models: common core
6. Categorical regression (multinomial distribution, multivariate GLM, nominal and ordinal models)
7. Linear mixed models (normal response, clustered data or repeated measurements)
8. Generalized mixed effects models (non-normal response, clustered data or repeated measurements)
9. Summing-up (this module)

Core of the course: regression

Main question: what is the effect of covariate(s) x on the response(s) y ?

Examples

- ▶ [M2] Munich rent index
- ▶ [M3] Mortality of beetles, infant respiratory disease, contraceptive use.
- ▶ [M4] Female crabs with satellites, smoking and lung cancer, time to blood coagulation, precipitation in Trondheim, treatment of breast cancer.
- ▶ [M6] Alligator food, mental health.
- ▶ [M7+8] Richness of species at beaches, sleep deprivation, trawl fishing.

The five ingredients

1. **Model specification:** an equation linking (conditional) mean of the response to and the explanatory variables, and a probability distribution for the response. We only consider responses from exponential family.
 - a. multiple linear regression model (normal response)
 - b. univariate generalized linear model (normal, binomial, Poisson, gamma)
 - c. multivariate generalized linear model (multinomial: nominal and ordinal)
 - d. linear mixed effect models (normal response, correlated within clusters)
 - e. generalized linear mixed models (binomial, Poisson)

2. **Likelihood** - used to estimate parameters (ML and a bit on REML): score function, Fisher information, Fisher scoring (IRWLS).
3. **Asymptotic distribution** of maximum likelihood estimators (multivariate normal) and tests (chisquared).
4. **Inference**: interpretation of results, plotting results, confidence intervals, hypothesis tests (Wald, LRT, score).
5. Checking the **adequacy of the model** (deviance, also residuals, qqplots - but very little focus in our course *outside the normal model*), **choose between models** (nested=LRT or AIC, not nested=AIC),

Understanding and comparing R print-outs

The print-outs below are from LM `lm`, GLM `glm`, mvGLM `vglm`, LMM `lmer` and GLMM `glmer`.

For the exam, you do not need to write code, but you still need to be able to interpret output!

Below we have fit a model to a data set, and then printed the summary of the model. For each of the print-outs you need to know (be able to identify and explain) every entry. In particular identify and explain:

- ▶ which model: model requirements
- ▶ how is the model fitted (versions of maximum likelihood)
- ▶ parameter estimates for β
- ▶ inference about the β : how to find CI and test hypotheses (which hypothesis is reported test statistic, and possibly p -value for)
- ▶ model fit (deviance, AIC, R-squared, F)

In addition, further inference can be made using `anova(fit1, fit2)`, `confint`, `residuals`, `fitted`, `AIC`, `ranef` and other functions we have worked with in the PL, IL and on the compulsory exercises.

MLR - multiple linear regression

```
fitLM=lm(rent~area+location+bath+kitchen+
          cheating,data=gamlss.data::rent99)
summary(fitLM)
```

```
##
```

```
## Call:
```

```
## lm(formula = rent ~ area + location + bath + kitchen + c
```

```
##      data = gamlss.data::rent99)
```

```
##
```

```
## Residuals:
```

```
##      Min        1Q   Median        3Q      Max
## -633.41  -89.17   -6.26   82.96 1000.76
```

```
##
```

```
## Coefficients:
```

```
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -21.9733    11.6549  -1.885   0.0595 .
## area         4.5788     0.1143  40.055 < 2e-16 ***
## location2    39.2602     5.4471   7.208 7.14e-13 ***
## location3   126.0575    16.8747   7.470 1.04e-13 ***
```



```
fitGLM=glm(rent~area,data=gamlss.data::rent99)
summary(fitGLM)
```

```
##
```

```
## Call:
```

```
## glm(formula = rent ~ area, data = gamlss.data::rent99)
```

```
##
```

```
## Coefficients:
```

```
##             Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept) 134.5922      8.6135   15.63  <2e-16 ***
```

```
## area         4.8215      0.1206   39.98  <2e-16 ***
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1
```

```
##
```

```
## (Dispersion parameter for gaussian family taken to be 25
```

```
##
```

```
##      Null deviance: 117945363  on 3081  degrees of freedom
```

```
## Residual deviance:  77646265  on 3080  degrees of freedom
```

```
## AIC: 39986
```

```
##
```

GLM - Binomial regression with logit-link

```
fitgrouped = glm(cbind(y, n - y) ~ ldose, family = "binomial")  
summary(fitgrouped)
```

```
##
```

```
## Call:
```

```
## glm(formula = cbind(y, n - y) ~ ldose, family = "binomial")
```

```
##
```

```
## Coefficients:
```

	Estimate	Std. Error	z value	Pr(> z)
## (Intercept)	-60.717	5.181	-11.72	<2e-16 ***
## ldose	34.270	2.912	11.77	<2e-16 ***

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
```

```
## (Dispersion parameter for binomial family taken to be 1)
```

```
##
```

```
## Null deviance: 284.202 on 7 degrees of freedom
```

```
## Residual deviance: 11.232 on 6 degrees of freedom
```

```
## AIC: 41.43
```

GLM - Poisson regression with log-link

```
crab = read.table("https://www.math.ntnu.no/emner/TMA4315/2018/
colnames(crab) = c("Obs", "C", "S", "W", "Wt", "Sa")
crab = crab[, -1]  #remove column with Obs
crab$C = as.factor(crab$C)
model3 = glm(Sa ~ W + C, family = poisson(link = log), data = crab)
summary(model3)
```

```
##
```

```
## Call:
```

```
## glm(formula = Sa ~ W + C, family = poisson(link = log),
##      contrasts = list(C = "contr.sum"))
```

```
##
```

```
## Coefficients:
```

	Estimate	Std. Error	z value	Pr(> z)	
## (Intercept)	-2.92089	0.56010	-5.215	1.84e-07	***
## W	0.14934	0.02084	7.166	7.73e-13	***
## C1	0.27085	0.11784	2.298	0.0215	*
## C2	0.07117	0.07296	0.975	0.3294	
## C3	-0.16551	0.09316	-1.777	0.0756	.

Categorical regression, nominal model

```
# data from Agresti (2015), section 6, with use of the VGAM
data = "http://www.stat.ufl.edu/~aa/glm/data/Alligators.dat"
ali = read.table(data, header = T)
attach(ali)
y.data = cbind(y2, y3, y4, y5, y1)
x.data = model.matrix(~size + factor(lake), data = ali)
library(VGAM)
# We fit a multinomial logit model with fish (y1) as the reference
fit.main = vglm(cbind(y2, y3, y4, y5, y1) ~ size + factor(lake),
               data = ali)
summary(fit.main)
pchisq(deviance(fit.main), df.residual(fit.main), lower.tail = FALSE)

##
## Call:
## vglm(formula = cbind(y2, y3, y4, y5, y1) ~ size + factor(lake),
##       family = multinomial, data = ali)
##
## Coefficients:
```

Categorical regression, ordinal model

```
# Read mental health data from the web:
```

```
fit.imp = VGAM::vglm(impair ~ life + ses, family = cumulative,  
summary(fit.imp))
```

```
##
```

```
## Call:
```

```
## VGAM::vglm(formula = impair ~ life + ses, family = cumulative,
```

```
##      data = mental)
```

```
##
```

```
## Coefficients:
```

```
##              Estimate Std. Error z value Pr(>|z|)
```

```
## (Intercept):1  -0.2819      0.6231  -0.452  0.65096
```

```
## (Intercept):2   1.2128      0.6511   1.863  0.06251 .
```

```
## (Intercept):3   2.2094      0.7171   3.081  0.00206 **
```

```
## life            -0.3189      0.1194  -2.670  0.00759 **
```

```
## ses              1.1112      0.6143   1.809  0.07045 .
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1
```

```
##
```

LMM - random intercept and slope

```
library(lme4)
fm1 <- lmer(Reaction ~ Days + (Days | Subject), sleepstudy)
summary(fm1)
```

Linear mixed model fit by REML ['lmerMod']
Formula: Reaction ~ Days + (Days | Subject)
Data: sleepstudy

REML criterion at convergence: 1743.6

Scaled residuals:
Min 1Q Median 3Q Max
-3.9536 -0.4634 0.0231 0.4634 5.1793

Random effects:
Groups Name Variance Std.Dev. Corr
Subject (Intercept) 612.10 24.741
Days 35.07 5.922 0.07
Residual 654.94 25.592

GLMM - random intercept and slope Poisson

```
RIKZ <- read.csv("http://faculty.concordia.ca/pperesne/BIO100/BIOL100Data/BIOL100Data.csv")
library(lme4)
fitRI = glmer(Richness ~ NAP + (1 + NAP | Beach), data = RIKZ)
summary(fitRI)
```

```
## Generalized linear mixed model fit by maximum likelihood (Eigenvalue  
## Approximation) [glmerMod]  
## Family: poisson ( log )  
## Formula: Richness ~ NAP + (1 + NAP | Beach)  
## Data: RIKZ  
##  
##           AIC           BIC    logLik deviance df.resid  
##      218.7       227.8   -104.4    208.7        40  
##  
## Scaled residuals:  
##           Min           1Q       Median           3Q           Max  
## -1.35846 -0.51129 -0.21846  0.09802  2.45384  
##  
## Random effects:
```

Exam and exam preparation

Relevant exams are found on the bottom of each module page.

▶ I will use these to help me write the exam!

Exam date: 05.12.2023 15:00

Revision sessions

Same times & places as lecture & interactive session, week before the exam

- ▶ 27th November 10:00 - 12:00: F4
- ▶ 28th November 10:00 - 12:00: G4

After TMA4315

What is next in the spring semester?

For the 4th year student

- ▶ TMA4250 Spatial statistics
- ▶ TMA4268 Statistical learning
- ▶ TMA4275 Survival analysis
- ▶ TMA4300 Computational statistics
- ▶ SMED8002 Epidemiology 2
- ▶ TDT4300 Data Warehousing and Data Mining
- ▶ TDT4173 Machine learning and case-based reasoning (Big overlap with TMA4268)

For the 5th year student

- ▶ Computational statistics 2 Phd course

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